LTM GSD Training

Active Directory Essentials

Section 1 — Fundamentals

Topic 1: Operating Systems and Types

An Operating System (OS) is software that manages a computer's hardware and provides a platform for applications to run. Without an OS, a computer cannot function.

Type	Purpose	Examples
Desktop OS	Used on everyday computers and laptops	Windows 10/11, macOS, Ubuntu
Server OS	Runs services for a network (files, AD, web)	Windows Server 2019/2022, Linux
Mobile OS	Used on smartphones and tablets	Android, iOS
Network OS	Runs inside routers and switches	Cisco IOS, Junos

AD Context: Active Directory runs on Windows Server OS. The computers that users log into run a Desktop OS (Windows 10/11).

Topic 2: Networking Basics — Key Components

A network is a group of devices connected together to share data and resources. Here are the four key components you need to know:

Component	What It Does
Host	Any device connected to a network — a PC, laptop, server, or printer. Every host has a unique IP address.
Switch	Connects multiple devices within the same network (e.g., within an office). Directs data only to the intended device, not to everyone.
Router	Connects different networks together (e.g., your office network to the internet). Decides the best path for data to travel.
Firewall	Acts as a security gatekeeper. Monitors incoming and outgoing traffic and blocks anything unauthorized or suspicious.

Simple Flow: In an office: All PCs connect to a Switch → Switch connects to a Router → Router connects to the Internet → Firewall protects the boundary.

Topic 3: Types of Networks

Type	Coverage	Example
LAN	Single building or office	Office computers connected via switch and cable
WAN	Across cities, countries, or globe	Internet; bank branches connected across India
MAN	A city or large campus	University network across multiple buildings
WLAN	Same as LAN but wireless (Wi-Fi)	Office or home Wi-Fi network
VPN	Secure tunnel over any network	Employee working from home accessing office systems

> **AD Context:** Active Directory operates within a LAN. For remote users or branch offices, a VPN is used so their machines can still reach the Domain Controller.

Topic 4: IP Addressing Overview

Every device on a network needs a unique address to send and receive data. This is called an IP Address — similar to a home address for your computer.

IPv4 vs IPv6

Feature	Details
IPv4	32-bit address. Written as four numbers separated by dots. Example: 192.168.1.10. Supports ~4.3 billion addresses. Still the most widely used.
IPv6	128-bit address. Written in hexadecimal with colons. Example: 2001:db8::1. Created because IPv4 addresses were running out.

Public IP vs Private IP

Type	Description
Public IP	Unique on the internet. Assigned by your ISP. This is how your organization is identified on the internet. Example: 203.122.45.10
Private IP	Used only inside a local network. Not visible on the internet. Common ranges: 10.x.x.x, 172.16.x.x, 192.168.x.x

NAT — Network Address Translation

Since devices inside an office use private IPs, NAT (running on the router) translates private IPs to the organization's single public IP when traffic goes out to the internet — and back again when the response returns.

> **Analogy:** NAT is like a company's single main phone number. Internally everyone has extensions (private IPs), but all outgoing calls show the main number (public IP).

Topic 5: IPv4 Classes

IPv4 addresses are grouped into classes based on the size of the network and the number of hosts they are designed for. Classes A, B, and C are used in practice.

Class	Range	Used For
A	1.0.0.0 – 126.255.255.255	Network bits: 8 | Host bits: 24 Very large networks → ~16 million hosts per network
B	128.0.0.0 – 191.255.255.255	Network bits: 16 | Host bits: 16 Medium organizations → ~65,000 hosts per network.
C	192.0.0.0 – 223.255.255.255	Network bits: 24 | Host bits: 8 Small offices and homes → up to 254 hosts per network
D	224.0.0.0 – 239.255.255.255	Multicast — not used for regular hosts
E	240.0.0.0 – 255.255.255.255	Experimental — reserved, not used in production

> **Special Address:** 127.0.0.1 is the loopback address. It always refers to your own machine. Commonly called 'localhost'. Used to test if your network stack is working.

Topic 6: DHCP and the DORA Process

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses to devices when they connect to the network. Without DHCP, an IT admin would have to manually configure the IP on every single device.

The DORA Process

When a device joins the network, it goes through four steps to get an IP address:

Step	Name	What Happens
D	Discover	Device broadcasts to the network: 'I need an IP address. Is there a DHCP server?'
O	Offer	DHCP server responds: 'Here is an available IP address for you.'
R	Request	Device replies: 'I accept that IP address. Please confirm.'
A	Acknowledge	DHCP server confirms: 'IP is assigned. Lease starts now.'

Lease: The IP is assigned for a set period (e.g., 8 hours). When the lease expires, the device renews it. If the device leaves the network, the IP is returned to the pool for reuse.

Topic 7: DNS — Domain Name System

DNS (Domain Name System) translates human-readable names like www.google.com into IP addresses that computers understand, like 142.250.183.46. It works like a phone book for the internet — you look up a name and get a number.

Every time you open a website, send an email, or log into a domain, DNS is working in the background to find the correct address.

Common DNS Record Types

Record Type	Purpose
A Record	Maps a domain name to an IPv4 address
AAAA Record	Maps a domain name to an IPv6 address
CNAME	Creates an alias pointing to another domain name
MX Record	Directs email traffic to the correct mail server
PTR Record	Reverse lookup — resolves an IP address back to a name
SRV Record	Locates services on the network — used by AD to find Domain Controllers

AD Context: DNS is critical for Active Directory. Computers use DNS (specifically SRV records) to locate Domain Controllers. If DNS breaks, users cannot log into the domain — this is one of the most common AD issues you will encounter on the Service Desk.

Topic 8: What is a Domain?

A domain is a centrally managed group of computers, users, and resources that all share the same Active Directory database and security policies. It is identified by a unique DNS name, such as lti.local or technova.com.

All accounts, policies, and access permissions are managed from one place — the Domain Controller. A user created in the domain can log into any domain-joined computer using the same credentials.

Analogy: A domain is like a company office building. The security desk (Domain Controller) manages who gets in and what areas they can access. Every employee (user) has one ID card (account) that works on every door in the building.

Key Point	Description
Centralized Management	All users, computers, and policies are managed from the Domain Controller.
Single Login	One username and password works on any computer in the domain.
Consistent Policies	Group Policies are pushed automatically to all users and computers.
Security Boundary	Each domain manages its own security. External access requires explicit trust.

Topic 9: Domain vs Workgroup

In a Windows environment, computers can be set up in one of two ways — a Workgroup or a Domain. Understanding the difference is key before moving into Active Directory.

Feature	Workgroup	Domain
Management	No central management — each PC manages itself	Centrally managed by a Domain Controller
User Accounts	Stored locally on each individual PC	Stored centrally in Active Directory
Login	Separate login for each PC	One login works on any domain-joined PC
Policies	Set manually on each machine	Applied automatically via Group Policy (GPO)
Security	Inconsistent — each PC decides its own rules	Consistent and centrally enforced
Scale	Suitable for fewer than 10 PCs	Scales to thousands of users easily
Server Required?	No	Yes — Windows Server with AD DS role

Key Takeaway: Any organization with more than a handful of computers will use a Domain. As a Service Desk engineer at LTI, every environment you work in will be domain-based. Workgroups are rarely seen in enterprise settings.

Section 2 — Active Directory Core Concepts

Topic 1: Introduction to Active Directory

Active Directory (AD) is a directory service developed by Microsoft. It runs on Windows Server and is used to centrally manage users, computers, groups, and other resources in an organization's network.

In simple terms — Active Directory is the central identity and access management system for a Windows-based corporate network. It answers two key questions for every login: Who are you? and What are you allowed to access?

Key Features at a Glance

Feature	What It Means
Centralized Management	All users, computers, and policies are managed from one place — the Domain Controller.
Authentication	AD verifies who you are when you log in (checks username and password).
Authorization	AD controls what resources you can access after logging in.
Group Policy (GPO)	Rules and settings pushed automatically to users and computers across the network.
Scalability	Works for small offices with 10 users or large enterprises with 100,000+ users.
Replication	Multiple Domain Controllers stay in sync automatically — ensuring availability.

In a nutshell: When a user at LTM logs in to any office computer, Active Directory is what checks their credentials, applies their settings, and decides what they can access — all in seconds.

Topic 2: Domains, Trees and Forests

Active Directory uses a logical structure to organize and manage resources across an organization. This structure has three levels: Domain, Tree, and Forest.

Domain

A domain is the basic unit of Active Directory. It is a logical grouping of users, computers, and resources that share the same AD database and security policies. Every domain has a unique DNS name (e.g., technova.local) and is managed by at least one Domain Controller.

Tree

A tree is a collection of one or more domains that share a common DNS namespace. The first domain is the root domain and any domains created under it are child domains. All domains in a tree automatically trust each other through two-way transitive trust relationships.

Example: technova.local is the root. sales.technova.local and hr.technova.local are child domains — together they form a tree.

Forest

A forest is the highest-level container in Active Directory. It is a collection of one or more trees that share a common AD schema and Global Catalog. The forest is the ultimate security boundary in AD — trust between forests must be manually configured.

Level	Description	Key Point
Domain	Basic unit — group of users, computers, resources	Unique DNS name; managed by a Domain Controller
Tree	Collection of domains with shared DNS namespace	Automatic two-way trust between domains
Forest	Collection of trees; highest-level AD container	Shared schema and Global Catalog; security boundary

> **Analogy:** Think of it like a company structure. A Domain is one office. A Tree is all offices under the same brand. A Forest is the entire group of companies under one holding company.

Topic 3: Domain Controllers and Types

A Domain Controller (DC) is a Windows Server with the Active Directory Domain Services (AD DS) role installed. It is the core of every AD environment — storing the AD database and handling all authentication and authorization requests.

Main Functions of a Domain Controller

- Authentication — Verifies usernames and passwords when users log in.
- Authorization — Determines what resources a user or computer can access.
- Directory Storage — Stores all AD data in a database file called NTDS.dit.
- Policy Enforcement — Applies Group Policies to users and computers.
- Replication — Keeps AD data synchronized with other Domain Controllers.

Types of Domain Controllers

Type	Description
Primary Domain Controller (PDC)	The first DC created when a domain is set up. Holds the original writable copy of the AD database. Performs critical roles like time synchronization and password change coordination.
Additional Domain Controller (ADC)	A second (or more) writable DC added to the same domain. Provides redundancy — if the PDC fails, the ADC continues to authenticate users and serve the network.
Read-Only Domain Controller (RODC)	Contains a read-only copy of the AD database. Used in branch offices or locations with limited physical security. No changes can be made directly on an RODC.
Child Domain Controller (CDC)	Manages a child domain (a sub-domain under the main domain). Handles authentication and policies for its own domain while maintaining a trust with the parent domain.

The AD Database — NTDS.dit

Every Domain Controller stores its copy of the Active Directory database in a file called NTDS.dit. This file contains all objects — users, computers, groups, OUs, and their attributes. It is stored at: C:\Windows\NTDS\NTDS.dit

Global Catalog (GC)

The Global Catalog is a special role that can be enabled on any Domain Controller. A GC server stores a partial copy of all objects across every domain in the forest — not full details, but key attributes like name, email, and group membership.

- Allows users to search for objects anywhere in the forest.
- Required for user logins in multi-domain environments.
- Reduces replication traffic — only partial object data is replicated across domains.

> **Key Point:** In most environments, every Domain Controller is also a Global Catalog server. This is recommended best practice for performance and availability.

Topic 4: Authentication in Active Directory

Authentication is the process of verifying the identity of a user, computer, or service before granting access to resources. In Active Directory, authentication happens every time someone logs in — the Domain Controller checks who you are before deciding what you can access.

> **Authentication vs Authorization:** Authentication = Proving who you are. Authorization = What you are allowed to do. Authentication always happens first.

Types of Authentication in AD

Type	Description and Example
Interactive Authentication	User logs in directly at a domain-joined machine. Example: An employee sitting at their office PC and logging in with their domain credentials.
Network Authentication	User or service accesses a network resource without a local login. Example: Opening a shared folder on a file server from your workstation.
Service Authentication	One service or application authenticates to another. Example: A web application authenticating to a back-end database server.
Remote Authentication	Authentication performed over VPN or Remote Desktop. Example: A remote employee connecting via VPN and logging into corporate systems from home.

Authentication Protocols

Active Directory uses two main protocols for authentication:

Kerberos

Kerberos is the default and preferred authentication protocol in Active Directory since Windows 2000. It uses a ticket-based system — meaning your password is never sent across the network after the initial login.

- Used when both client and server are domain-joined and in the same forest.
- Provides mutual authentication — both the user and the server verify each other.
- Operates on Port 88 (TCP/UDP).
- Tickets have a limited validity period, reducing security risks.

NTLM (NT LAN Manager)

NTLM is the older authentication protocol. It is still supported but used only as a fallback when Kerberos is not available.

- Used for legacy systems, standalone (non-domain) machines, or cross-forest scenarios without a trust.
- Less secure than Kerberos — no mutual authentication.
- Still encountered in older environments and some specific application integrations.

LDAP (Lightweight Directory Access Protocol)

LDAP is not strictly an authentication protocol — it is used to query and modify directory information in AD. Many applications use LDAP to look up user details or authenticate users against AD.

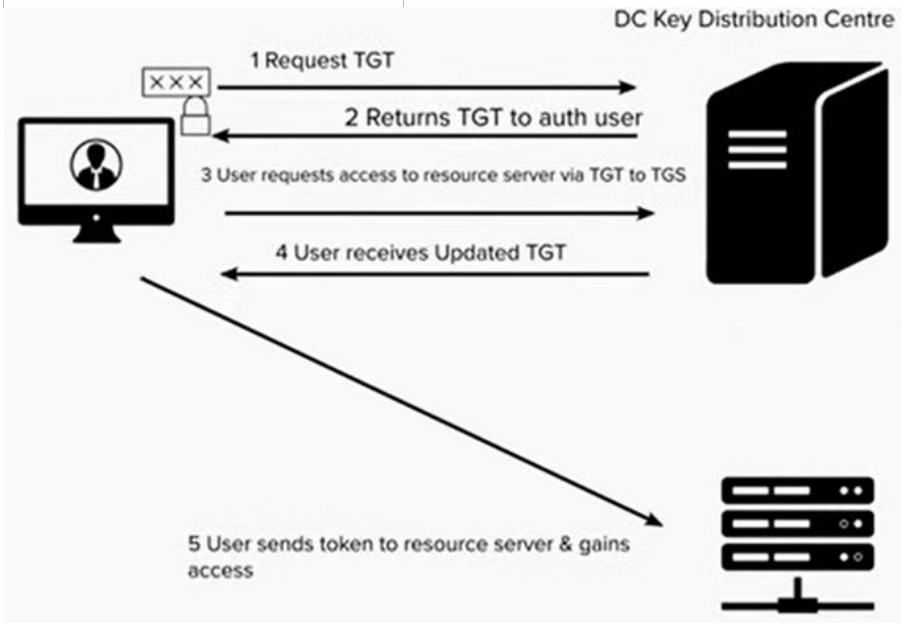
- Port 389 — standard LDAP (unencrypted).
- Port 636 — LDAPS (LDAP over SSL, encrypted). Always preferred.
- Example: An HR application querying AD via LDAP to retrieve all users in the Employees OU.

Protocol	Key Points
Kerberos	Default in AD. Ticket-based. Mutual authentication. Port 88. Most secure.
NTLM	Legacy fallback. Used when Kerberos is unavailable. Less secure. No mutual auth.
LDAP/LDAPS	Used by applications to query AD. Port 389 (plain) or 636 (encrypted).

How Kerberos Works — The TGT Process

When a user logs in, Kerberos issues tickets that are then used to access resources — without re-entering the password each time. This is managed by the Key Distribution Center (KDC), which runs on every Domain Controller.

Step	What Happens
1. User Logs In	User enters credentials at their workstation. The request is sent to the KDC on the Domain Controller.
2. TGT Issued	KDC verifies the credentials against AD. If valid, it issues a Ticket Granting Ticket (TGT) — encrypted and time-limited. The TGT contains the user's SID, group memberships, and validity period.
3. Service Access Requested	When the user tries to access a resource (e.g., a file server), the workstation sends the TGT back to the KDC requesting a Service Ticket for that specific resource.
4. Service Ticket Issued	The KDC issues a Service Ticket encrypted with the target server's key.
5. Access Granted	The workstation presents the Service Ticket to the resource server. The server validates it and grants access — no password re-entry needed.



TGT – Ticket granting Ticket

TGS – Ticket granting Service

Real-World Example: Priya logs into her PC in the morning — a TGT is issued. When she opens a shared folder on the file server, her TGT is used to get a Service Ticket for that server automatically. She never has to type her password again for the rest of the day.

Service Desk Context: Most login failures on the Service Desk involve Kerberos or NTLM issues. Common causes include incorrect system time (Kerberos requires clocks to be within 5 minutes of the DC), account lockouts, or expired tickets. If a user cannot access a resource, checking whether Kerberos or NTLM is in use can help narrow down the root cause.

Topic 5: Active Directory Structure and Hierarchy

Active Directory organizes all network resources in a structured, hierarchical model. This makes it easy to manage, apply policies, and delegate control across large organizations.

Organizational Units (OUs)

An Organizational Unit (OU) is a container within a domain used to group objects like users, computers, and groups. OUs mirror the organization's structure — by department, location, or function.

- Group Policy Objects (GPOs) can be linked to specific OUs.
- Administrative control can be delegated at the OU level.
- Example OUs: Finance, HR, IT, Sales — each with their own policies and admin rights.

AD Objects

Everything stored in Active Directory is an object. Each object represents something real in the network.

Object Type	Represents
User Object	An individual user account (e.g., priya.sharma@technova.local)
Computer Object	A domain-joined computer or server
Group Object	A collection of users or computers for easier permission management
OU	A container to organize and manage other objects
Printer / Share	Network-accessible printers and shared folders

Default Containers vs OUs

When AD is first set up, it creates default containers like Users and Computers. These are different from OUs — default containers cannot have Group Policies linked to them. Best practice is to move objects into custom OUs for better management.

Sites and Subnets

A Site in AD represents a physical location (e.g., Mumbai office, Delhi office). Sites are mapped to IP subnets, and AD uses this mapping to direct users to the nearest Domain Controller for faster logins and efficient replication.

> **Example:** If a user in the Mumbai office logs in, AD uses Sites and Subnets to authenticate them against the Mumbai DC — not one in Delhi — making the login faster.

Topic 6: Active Directory Server Roles

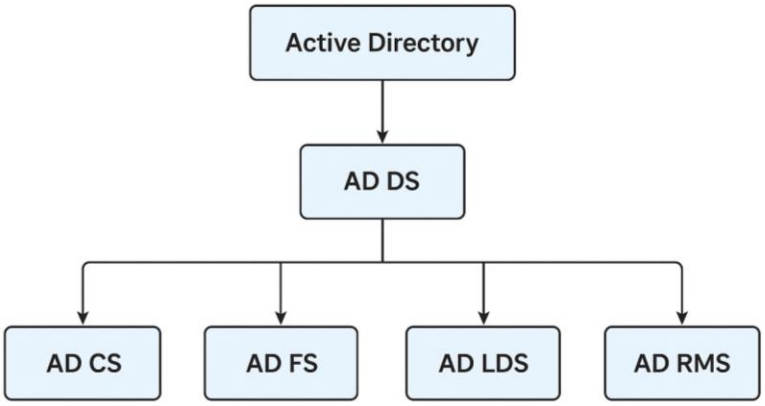
Active Directory is not a single service — it is a collection of roles that can be installed on Windows Server. Each role serves a specific purpose.

Role	Purpose and Key Use
AD DS — Active Directory Domain Services	The core AD role. Stores and manages all identity information — users, computers, groups, and policies. When installed, the server becomes a Domain Controller. This is what most people mean when they say 'Active Directory'.
AD CS — Active Directory Certificate Services	Provides Public Key Infrastructure (PKI). Issues and manages digital certificates used for encryption, authentication, and secure communication (e.g., HTTPS, smart card logins, VPN).
AD FS — Active Directory Federation Services	Enables Single Sign-On (SSO) across different organizations or cloud applications. A user logs in once with their AD credentials and gets access to Microsoft 365, Salesforce, or partner applications without logging in again.
AD LDS — Active Directory Lightweight Directory Services	A lightweight, standalone directory service. Used by applications that need a directory database but don't need full domain integration. Runs independently of AD DS.

Role	Purpose and Key Use
AD RMS — Active Directory Rights Management Services	Protects sensitive documents and emails by attaching usage policies directly to the content. Prevents unauthorized copying, printing, or forwarding — even if the file is shared outside the organization.

Service Desk Focus: As a Service Desk engineer, your primary work will involve AD DS. AD FS is relevant when users report SSO issues with Microsoft 365 or cloud apps. AD CS comes up when certificate errors appear.

Server Roles in Active Directory



Topic 7: Trust Relationships

In environments with multiple domains or forests, trust relationships allow users from one domain to access resources in another. Without a trust, domains are completely isolated from each other.

Direction of Trust

Type	How It Works
One-Way Trust	Domain A trusts Domain B — users from B can access resources in A, but not the other way around.
Two-Way Trust	Both domains trust each other — users from either domain can access resources in both. This is the default within a forest.

Transitivity of Trust

Type	How It Works
Transitive Trust	Trust extends automatically beyond two domains. If A trusts B and B trusts C, then A automatically trusts C. Default behavior within a forest.
Non-Transitive Trust	Trust exists only between the two specific domains. Does not extend further. Typically used between separate forests or external domains.

Common Trust Types

Trust Type	Direction	Transitive?	When Used
Parent-Child	Two-way	Yes	Auto-created when a child domain is added to a tree
Tree-Root	Two-way	Yes	Auto-created between tree roots in the same forest
External	One or Two-way	No	Manual — between domains in different forests
Forest	One or Two-way	Yes	Manual — collaboration between two separate forests
Shortcut	One or Two-way	Yes	Manual — speeds up logins in large, complex forests
Realm	One or Two-way	Either	Connecting AD to non-Windows Kerberos environments

Service Desk Context: You won't configure trusts day-to-day, but you need to understand them when users report they cannot access resources in another domain. The likely cause is a missing or broken trust relationship.

Topic 8: Active Directory Sites and Replication

What is an AD Site?

An AD Site represents a physical location or network segment with high-speed, reliable connectivity (typically a LAN). Sites are used to control how and when AD data replicates between Domain Controllers in different locations.

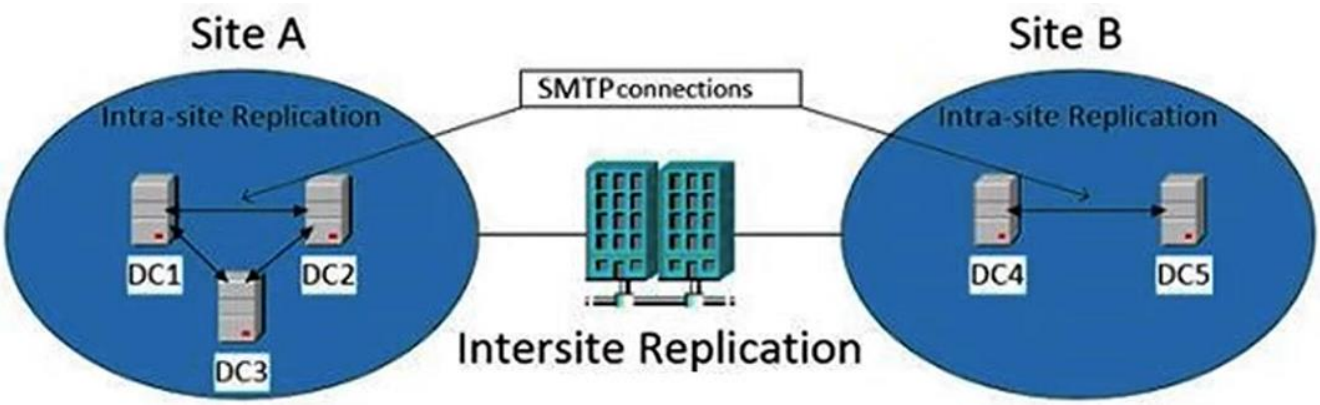
Component	Purpose
Site	Logical representation of a physical location (e.g., Mumbai office)
Subnet	IP range associated with a site (e.g., 192.168.1.0/24)
Site Link	Defines the replication path and schedule between two sites
Bridgehead Server	The designated DC in a site that handles replication traffic to/from other sites

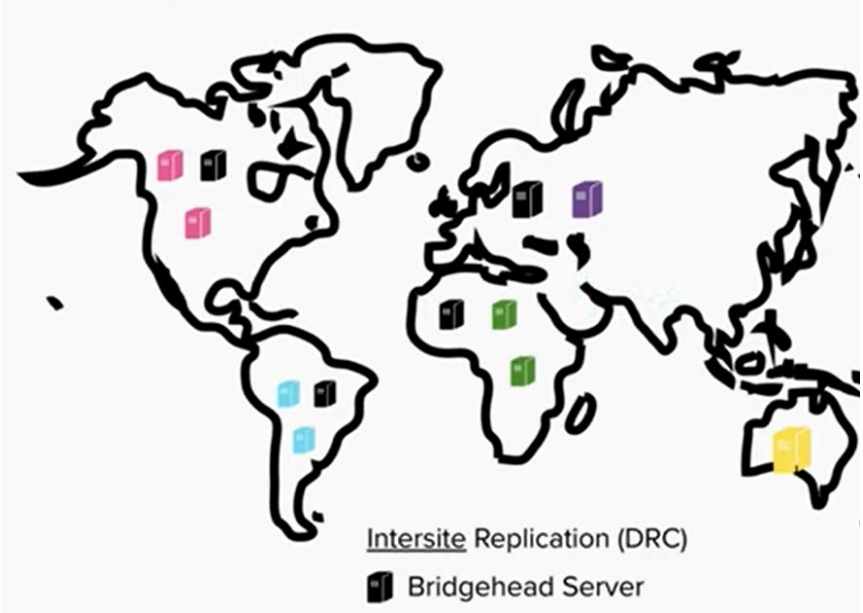
What is Replication?

Replication is the process of keeping all Domain Controllers synchronized. When a change is made on one DC (e.g., a new user is created or a password is reset), that change is automatically copied to all other DCs.

Types of Replication

Type	Description
Intra-Site Replication	Between DCs in the same site (same office). Uses fast LAN. Changes replicate almost instantly.
Inter-Site Replication	Between DCs in different sites (different offices). Scheduled to reduce WAN bandwidth usage. Data is compressed. Handled by the Bridgehead Server.





KCC — Knowledge Consistency Checker

The KCC is an automatic internal AD process that builds and maintains the replication topology. It ensures every DC can reach every other DC and automatically adjusts if a DC goes offline.

> **Service Desk Context:** Replication issues can cause problems like a password reset working on one DC but not another. If a user says their new password works on one computer but not others, replication delay is a common cause.

Topic 9: FSMO Roles

Active Directory uses a multi-master replication model — meaning changes can be made on any Domain Controller. However, certain critical operations must be handled by only one DC at a time to avoid conflicts. These are called FSMO Roles (Flexible Single Master Operations).

There are five FSMO roles in total — two apply to the entire forest, three apply per domain.

Forest-Wide Roles (One per Forest)

Role	Responsibility
Schema Master	Controls all changes to the AD schema (the blueprint defining what objects and attributes exist). Only this DC can update the schema.
Domain Naming Master	Controls adding or removing domains in the forest. Ensures every domain name is unique across the forest.

Domain-Wide Roles (One per Domain)

Role	Responsibility
RID Master	Allocates blocks of Relative IDs (RIDs) to other DCs. RIDs are used to create unique Security Identifiers (SIDs) for every new object created in AD.
Infrastructure Master	Tracks and updates references to objects from other domains — for example, if a user from another domain is added to a local group and later renamed.
PDC Emulator	The most active FSMO role. Handles password change synchronization, account lockouts, time synchronization for the domain, and Group Policy updates.

FSMO Role	Scope	If Unavailable	Most Critical?
Schema Master	Forest	Schema changes blocked; normal AD works fine	Low day-to-day impact
Domain Naming Master	Forest	Cannot add/remove domains; existing ones work fine	Low day-to-day impact
RID Master	Domain	New object creation fails once local RID pool runs out	Medium
Infrastructure Master	Domain	Cross-domain group membership may be stale	Medium
PDC Emulator	Domain	Password sync delays, lockout issues, time drift	High — most impactful

Service Desk Context: The PDC Emulator is the most relevant FSMO role for Service Desk. If users report account lockouts not clearing, password resets not taking effect immediately, or time sync issues — the PDC Emulator is the first place to investigate.

Topic 10: Group Policy

Group Policy is a feature in Active Directory that allows administrators to define and enforce settings, rules, and configurations for users and computers across the domain — automatically, without touching each machine individually.

What is a GPO?

A Group Policy Object (GPO) is a collection of settings. Once created, a GPO is linked to a Site, Domain, or OU — and its settings are automatically applied to all users and computers within that container.

GPO Application Order

When multiple GPOs apply to the same user or computer, they are processed in this order — and the last one applied wins (highest priority):

- 1. Local GPO — Settings on the individual computer itself.
- 2. Site GPO — Applied to all objects in an AD site.
- 3. Domain GPO — Applied to all objects in the domain.
- 4. OU GPO — Applied to objects in a specific OU. This takes highest priority.

Memory Aid: Remember the order as L-S-D-O (Local → Site → Domain → OU). The OU always wins in a conflict unless a higher-level GPO is set to Enforced.

Types of GPO Settings

Category	Applies To and Examples
Computer Configuration	Applied when the computer starts — regardless of who logs in. Examples: disable USB ports, configure Windows Firewall, install software.
User Configuration	Applied when a user logs in — regardless of which computer. Examples: set desktop wallpaper, redirect My Documents to a network share, restrict Control Panel access.

Common Uses of Group Policy

- Enforce password complexity and account lockout rules.
- Block access to USB drives, Task Manager, or Control Panel.
- Automatically install or remove software across all machines.
- Redirect user folders (Documents, Desktop) to a network location.
- Configure Windows Update and security settings centrally.

Key GPO Tools

Tool	Purpose
GPMC (Group Policy Management Console)	Main tool for creating, editing, linking, and managing GPOs
gpedit.msc	Local Group Policy Editor — view or edit settings on a single machine
gpupdate /force	Command to force an immediate refresh of all Group Policies on a machine
gpresult /r	Shows which GPOs are currently applied to the logged-in user and computer
RSoP	Resultant Set of Policy — shows the combined effect of all GPOs applied to a user or computer

Topic 11: AD Objects — Users and Groups

User Accounts

A user account is an AD object that allows a person to log on to the domain and access network resources. Each user account has a unique username (called sAMAccountName or UPN) and a password.

Account Type	Used By	Purpose
Standard User Account	End users	Log into computers, access email, shared drives, and applications.
Administrative Account	IT Administrators	Perform admin tasks — creating users, managing policies, server configuration.
Service Account	Applications / Services	Run background services (e.g., SQL Server, backup agents). Usually has no interactive login and a non-expiring password.

Groups

A group is a collection of user or computer accounts. Instead of assigning permissions to individuals, administrators assign permissions to groups — making access management much easier.

Group Types

Type	Purpose
Security Group	Used to assign permissions to resources like shared folders, printers, and applications. This is the most commonly used group type.
Distribution Group	Used for email distribution lists only (e.g., in Microsoft Exchange). Cannot be used to assign permissions.

Group Scope

Group scope defines where in the AD hierarchy the group can be used and who can be a member.

Scope	Members Can Be From	Can Be Used In	Common Use
Domain Local	Any domain in the forest	Same domain only	Assign permissions to resources in the local domain
Global	Same domain only	Any domain in the forest	Represent department-level users (e.g., HR team)
Universal	Any domain in the forest	Any domain in the forest	Enterprise-wide access or cross-domain resource sharing

Best Practice — AGDLP: The recommended group nesting model is: Accounts → Global Group → Domain Local Group → Permissions. Add user accounts to Global groups (by department), nest those into Domain Local groups (by resource), and assign permissions to the Domain Local groups.

Topic 12: Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) is a method of managing access permissions based on job roles rather than individual users. Instead of giving each person their own unique set of permissions, you define roles and assign permissions to those roles.

How RBAC Works in AD

- Identify roles in the organization — e.g., HR Executive, IT Support, Finance Analyst.
- Create an AD Security Group for each role.
- Assign the required permissions to each group.
- Add users to the group that matches their job role.
- When a user changes roles, simply move them to the new group — permissions update automatically.

RBAC Step	Example
Define Role	IT Support Engineer
Create AD Group	IT_Support_Access
Assign Permissions	Access to MECM console, Admin shares, Helpdesk tool
Add User to Group	User 'Amit' added to IT_Support_Access → gets all IT Support permissions instantly
User Leaves Role	Amit moved to Finance team → removed from IT_Support_Access → loses all IT access immediately

Service Desk Responsibilities under RBAC

- Add or remove users from AD groups based on HR requests or access tickets.
- Verify a user's current group membership when troubleshooting access issues.
- Never assign direct permissions to individual user accounts — always use groups.
- Escalate any requests that require creating new groups or modifying permissions to senior admins.

Why RBAC Matters: RBAC ensures the principle of least privilege — users only have access to what they need for their job. It also makes audits, access reviews, and offboarding much faster and cleaner.

Topic 13: Hybrid Active Directory

Most modern organizations use both on-premises Active Directory (running on Windows Server in their own data center) and cloud-based services like Microsoft 365. Hybrid AD connects these two environments so users have one identity for both.

What is Hybrid AD?

Hybrid AD = On-premises AD DS + Azure Active Directory (Azure AD), connected and synchronized through a tool called Azure AD Connect.

Users created in on-premises AD are automatically synced to Azure AD. This means the same username and password works for logging into the office PC and accessing cloud services like Outlook, Teams, and SharePoint Online.

Key Components

Component	Role
On-Premises AD DS	The local Active Directory running on Windows Server — source of truth for user identities.
Azure AD	Microsoft's cloud-based identity platform. Used for authenticating access to Microsoft 365 and cloud apps.
Azure AD Connect	The sync tool installed on-premises. It continuously synchronizes user accounts and password hashes from on-prem AD to Azure AD.
Single Sign-On (SSO)	Once synced, users log in once (on-prem) and seamlessly access all cloud applications without being prompted to log in again.

Hybrid Identity Authentication Options

Method	How It Works	Authentication Happens At
Password Hash Sync (PHS)	Password hashes are synced to Azure AD. Azure AD handles authentication.	Cloud (Azure AD)
Pass-Through Auth (PTA)	Azure AD passes the login request to an on-prem agent, which validates against local AD.	On-Premises AD
Federation (AD FS)	A separate AD FS server handles all authentication. Most complex to maintain.	On-Prem via AD FS

Most Common Setup: The majority of organizations today use Password Hash Synchronization (PHS) — it is the simplest, most reliable option and requires the least maintenance.

Topic 14: Active Directory Backup and Recovery

Active Directory stores the most critical data in any Windows environment — all user accounts, computer accounts, group memberships, and Group Policies. If this data is lost or corrupted, the entire domain can be affected.

What is Backed Up?

A System State Backup captures everything needed to restore a Domain Controller:

- NTDS.dit — The AD database file containing all objects and attributes.
- SYSVOL — Contains Group Policy files and logon scripts.
- Registry — System configuration data.
- Boot files and system files.

Tool Used: System State Backups on Windows Server are taken using Windows Server Backup (wbadmin). Service Desk staff should verify that scheduled backups are completing successfully.

Types of AD Restore

Restore Type	When to Use It
Non-Authoritative Restore	Used when a DC has failed and needs to be rebuilt. The DC is restored from backup and then re-syncs with other healthy DCs to get the latest changes. Most common scenario.
Authoritative Restore	Used when an object (like a user account or OU) was accidentally deleted and needs to be recovered. The restored object is marked as authoritative so other DCs do not overwrite it during replication.

Service Desk Role in Backup and Recovery

- Monitor backup job status and raise alerts if backups fail.
- Do not perform AD restores independently — always escalate to senior AD admins.
- If a user account or OU is accidentally deleted, raise a P1 ticket immediately — the sooner it is escalated, the easier the recovery.
- Know the difference between Non-Authoritative and Authoritative restore so you can describe the issue accurately to the escalation team.